AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Anish M. Naik
Roll No. :	79
Experiment No.:	1
Title of the Experiment:	Perform Exploratory Statistical Analysis of a Real-World Dataset (Case Study)
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Signature :

Date :

Aim: To perform exploratory statistical analysis of the Pima Indians Diabetes Dataset by examining its structure, variable types, data quality, statistical characteristics, and important patterns.

Objectives

After completing this experiment, the student will be able to:

1. Apply exploratory data analysis techniques to understand the structure and characteristics of a real-world healthcare dataset.
2. Interpret the statistical characteristics, data quality issues, and important patterns identified during the analysis.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations, 8 input attributes, and one binary outcome variable.

Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Examine the dataset dimensions, column names, data types, and sample records.
3. Classify the variables as numerical, categorical, or binary.
4. Generate a statistical summary of the dataset.
5. Check for missing or invalid values.
6. Identify potential outliers using suitable statistical techniques and visualizations.
7. Generate appropriate visualizations to understand the data.
8. Record and interpret important observations obtained from the exploratory analysis.

Description of the Experiment

Exploratory Data Analysis (EDA) is the initial process of understanding a dataset before applying statistical inference or machine learning techniques.

The analysis follows: Raw Dataset, Data Inspection, Variable Identification, Data Quality, Analysis, Statistical Summary, Visualization and Interpretation

The Pima Diabetes Dataset is explored to understand its structure, data quality, variable distributions, and important characteristics before developing predictive models.

Detailed Description of Statistical techniques:

Descriptive Statistics

Statistical measures such as the following are used to summarize numerical variables:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard Deviation

Missing Value Analysis

The dataset is examined for missing or invalid values.

For example, zero values in certain medical attributes such as Glucose, BloodPressure, SkinThickness, Insulin, and BMI may require investigation because a zero value may represent missing or invalid data rather than a genuine medical measurement.

Outlier Analysis

Potentially unusual observations are identified using statistical summaries and visualizations such as boxplots.

An outlier should be investigated before deciding whether it should be removed or retained.

Data Visualization

Suitable visualizations are generated to understand:

- Variable distributions
- Frequency of diabetes outcomes Relationships between variables
- Potential outliers Examples include:
- Histograms
- Boxplots
- Bar charts
- Scatter plots

CODE:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns


df = pd.read_csv("diabetes.csv")

print("Shape of Dataset:", df.shape)

print("\nColumns:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nFirst 5 Records:")
print(df.head())
```

```
print("\nNumerical Variables:")
for col in df.columns:
    if df[col].nunique() > 2:
        print("-", col)

print("\nBinary Variable:")
print("Outcome")


print("\nStatistical Summary")
print(df.describe())

print("\nMedian")
print(df.median())

print("\nVariance")
print(df.var())

print("\nStandard Deviation")
print(df.std())

print("\nMissing Values")
print(df.isnull().sum())

print("\nZero Values in Important Columns")

columns = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]

for col in columns:
    print(col, ":", (df[col] == 0).sum())

print("\nPossible Outliers (IQR Method)")

for col in df.columns[:-1]:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
```

```
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower) | (df[col] > upper)]

    print(col, ":", len(outliers))

df.hist(figsize=(12,10))
plt.show()

plt.figure(figsize=(12,8))

for i, col in enumerate(df.columns[:-1]):
    plt.subplot(3,3,i+1)
    sns.boxplot(y=df[col])

plt.tight_layout()
plt.show()

sns.countplot(x="Outcome", data=df)
plt.title("Diabetes Outcome")
plt.show()

plt.figure(figsize=(8,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.show()

sns.scatterplot(data=df, x="Glucose", y="BMI", hue="Outcome")
plt.show()
```

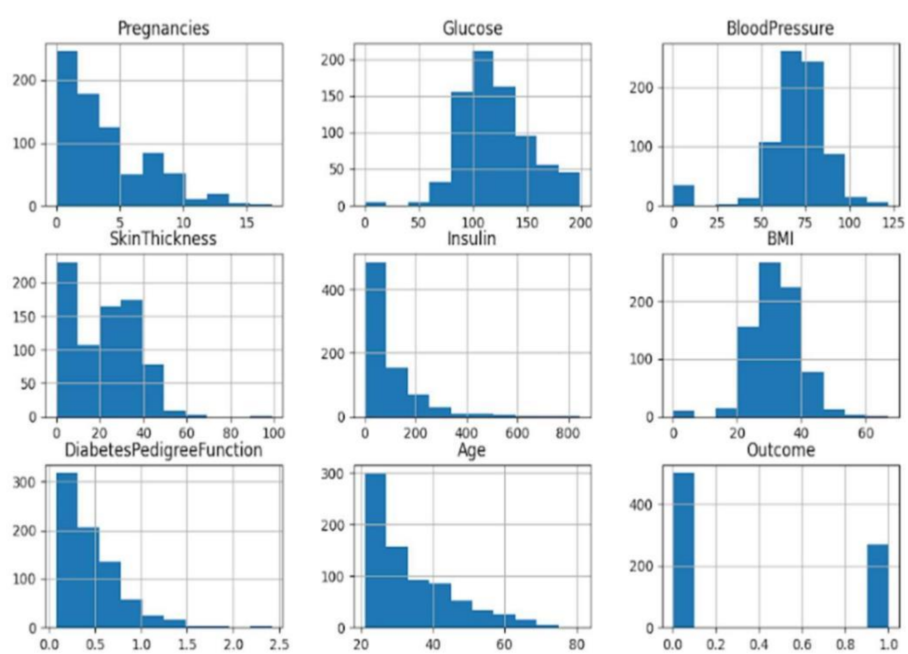
Output:-

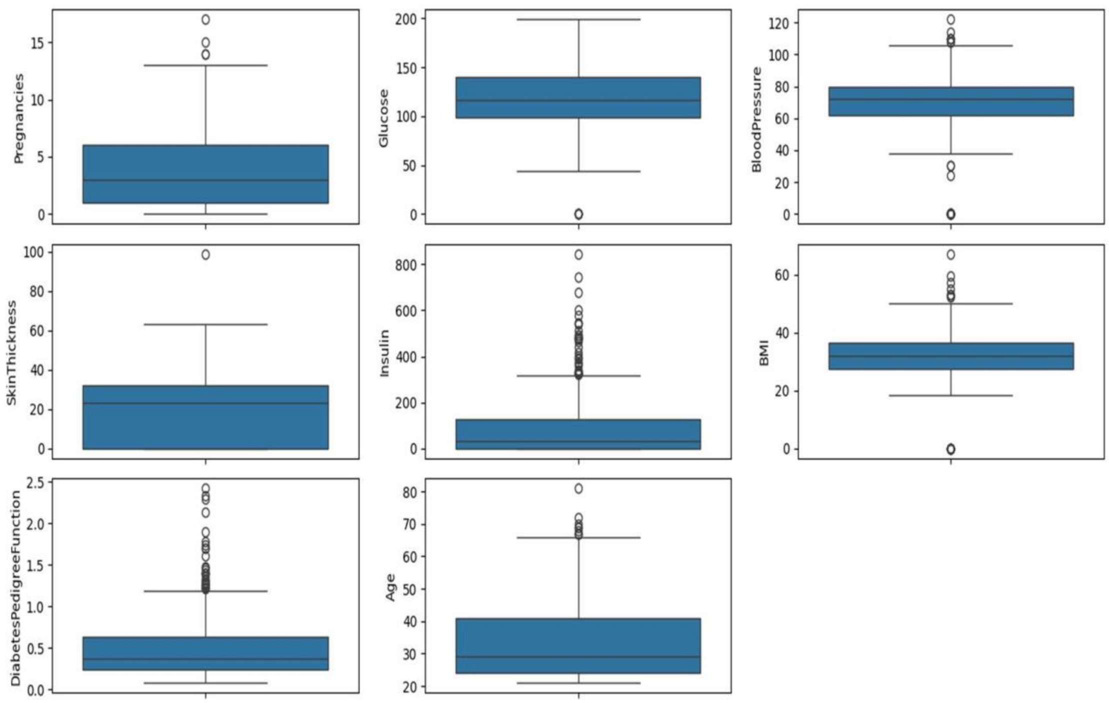
```
Shape of Dataset: (768, 9)

Columns:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='object')

Data Types:
Pregnancies                   int64
Glucose                       int64
BloodPressure                 int64
SkinThickness                 int64
Insulin                       int64
BMI                         float64
DiabetesPedigreeFunction    float64
Age                           int64
Outcome                       int64
dtype: object

First 5 Records:
   Pregnancies  Glucose  BloodPressure  ...  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72  ...                     0.627   50        1
1            1       85             66  ...                     0.351   31        0
2            8      183             64  ...                     0.672   32        1
3            1       89             66  ...                     0.167   21        0
4            0      137             40  ...                     2.288   33        1

[5 rows x 9 columns]

Numerical Variables:
- Pregnancies
- Glucose
- BloodPressure
```

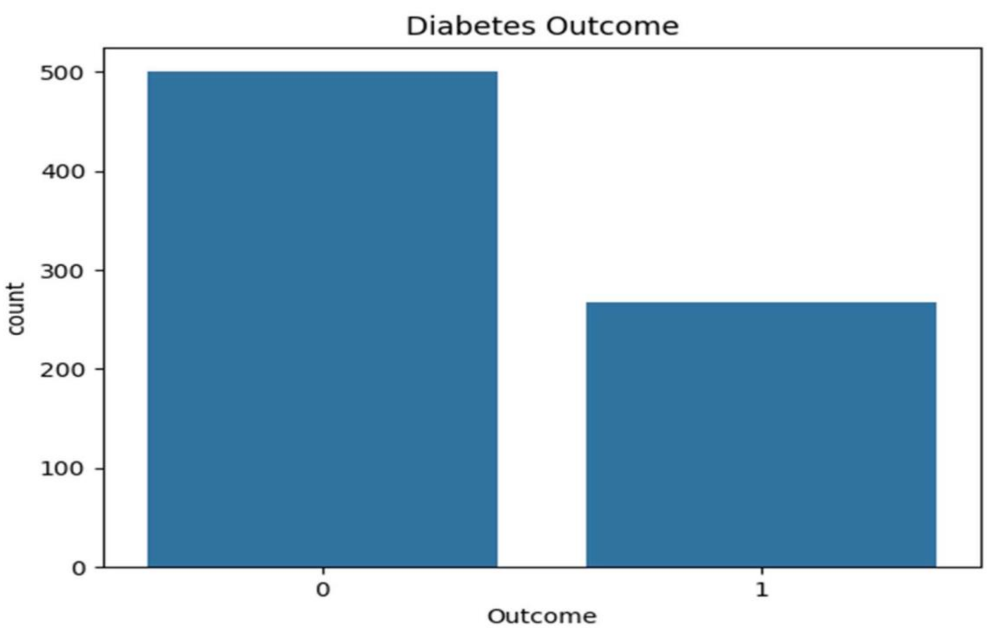
```
Numerical Variables:
- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age

Binary Variable:
Outcome

Statistical Summary
       Pregnancies     Glucose  BloodPressure  ...  DiabetesPedigreeFunction         Age     Outcome
count   768.000000  768.000000     768.000000  ...                768.000000  768.000000  768.000000
mean      3.845052  120.894531      69.105469  ...                  0.471876   33.240885    0.348958
std       3.369578   31.972618      19.355807  ...                  0.331329   11.760232    0.476951
min       0.000000    0.000000       0.000000  ...                  0.078000   21.000000    0.000000
25%       1.000000   99.000000      62.000000  ...                  0.243750   24.000000    0.000000
50%       3.000000  117.000000      72.000000  ...                  0.372500   29.000000    0.000000
75%       6.000000  140.250000      80.000000  ...                  0.626250   41.000000    1.000000
max      17.000000  199.000000     122.000000  ...                  2.420000   81.000000    1.000000

[8 rows x 9 columns]

Median
Pregnancies                   3.0000
Glucose                     117.0000
BloodPressure                72.0000
SkinThickness                23.0000
Insulin                      30.5000
BMI                          32.0000
DiabetesPedigreeFunction      0.3725
Age                          29.0000
```

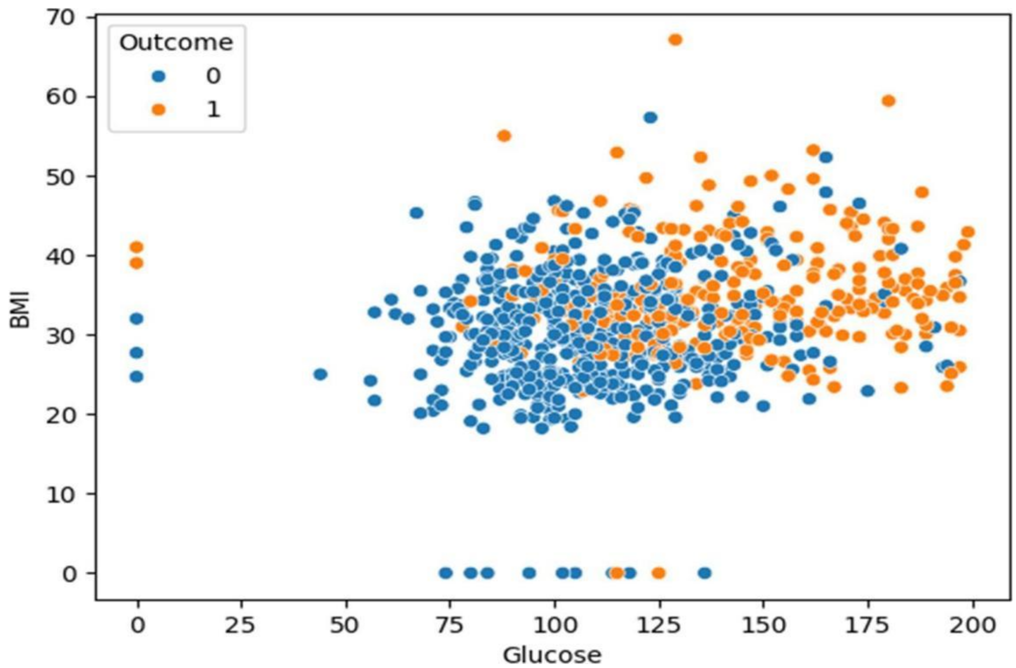
```
DiabetesPedigreeFunction      0.3725
Age                          29.0000
Outcome                       0.0000
dtype: float64

Variance
Pregnancies                    11.354056
Glucose                      1022.248314
BloodPressure                 374.647271
SkinThickness                 254.473245
Insulin                     13281.180078
BMI                            62.159984
DiabetesPedigreeFunction        0.109779
Age                           138.303046
Outcome                         0.227483
dtype: float64

Standard Deviation
Pregnancies                   3.369578
Glucose                      31.972618
BloodPressure                19.355807
SkinThickness                15.952218
Insulin                     115.244002
BMI                           7.884160
DiabetesPedigreeFunction      0.331329
Age                          11.760232
Outcome                       0.476951
dtype: float64

Missing Values
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
```

```
Insulin                     115.244002
BMI                           7.884160
DiabetesPedigreeFunction      0.331329
Age                          11.760232
Outcome                       0.476951
dtype: float64

Missing Values
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Zero Values in Important Columns
Glucose : 5
BloodPressure : 35
SkinThickness : 227
Insulin : 374
BMI : 11

Possible Outliers (IQR Method)
Pregnancies : 4
Glucose : 5
BloodPressure : 45
SkinThickness : 1
Insulin : 34
BMI : 19
DiabetesPedigreeFunction : 29
Age : 9
```




Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age


Diabetes Outcome
count
Outcome
0
1
0
100
200
300
400
500



Conclusion

Exploratory statistical analysis was performed on the Pima Indians Diabetes Dataset to understand its structure, variable characteristics, data quality, and important patterns. The analysis provides the foundation for further statistical analysis and machine learning model development.

Questions to be Answered by Students

1. Which variables in the dataset contain zero or potentially invalid values that require investigation?

The following variables contain zero values that may represent missing or invalid medical measurements and should be investigated:

- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI

These variables cannot normally have a value of zero in a clinical setting, so such values are often treated as missing data during preprocessing

2. Which variable contains potential outliers based on the analysis?

The variables with potential outliers are:

- Insulin (most significant outliers)
- BMI
- DiabetesPedigreeFunction
- Pregnancies

These outliers were identified using boxplots and the Interquartile Range (IQR) method

3. What are the important patterns observed between the input variables and the Outcome variable?

The important patterns observed are:

- Glucose has the strongest positive relationship with the Outcome variable; diabetic patients generally have higher glucose levels.
- BMI tends to be higher among patients with diabetes.
- Age shows a moderate positive association with diabetes, with older individuals more likely to have diabetes.
- Pregnancies are generally higher among diabetic patients.
- The correlation matrix indicates that Glucose is the most influential predictor, followed by BMI and Age.

4. State two important insights obtained from the exploratory analysis.

1. Glucose is the strongest predictor of diabetes, with diabetic patients having noticeably higher glucose levels than non-diabetic patients.
2. The dataset contains invalid zero values and several outliers, indicating that data preprocessing (handling missing values and evaluating outliers) is necessary before performing statistical analysis or building machine learning models.